

Create index.html

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

Create a Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Create a deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
```

```

selector:
  matchLabels:
    app: html-demo
template:
  metadata:
    labels:
      app: html-demo
  spec:
    containers:
      - name: html-demo
        image: <<dockerhub_username>>/html-demo:latest
        ports:
          - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010

```

Create Pipeline

```

pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "<<dockerhubusername>>/html-demo"
    }

    stages {

```

```
stage('Clone Code') {  
    steps {  
        git branch: 'main',  
            url: '<<UserGITHUB repository>>'  
    }  
}
```

```
stage('Build Docker Image') {  
    steps {  
        bat 'docker build -t %DOCKER_IMAGE% .'  
    }  
}
```

```
stage('Push Image') {  
    steps {  
        withCredentials([usernamePassword(credentialsId: 'dockerhub',  
usernameVariable: 'USER', passwordVariable: 'PASS')]) {  
            bat 'docker login -u %USER% -p %PASS%'  
            bat 'docker push %DOCKER_IMAGE%'  
        }  
    }  
}
```

```
stage('Deploy to Kubernetes') {  
    steps {  
        withCredentials([file(credentialsId: 'kuberconfig', variable:  
'KUBECONFIG')]) {  
            bat '''
```

```
set KUBECONFIG=%KUBECONFIG%
```

```
kubectrl apply -f deployment.yaml --validate=false
```

```
'''
```

```
    }
```

```
  }
```

```
  }
```

```
}
```

```
}
```